CSA1718 : Artificial Intelligence

Name : D. Jeshurun Silas

Reg No : 192525324

10.A* Search:

```
import heapq

graph = {
    "A": [("B", 1), ("C", 4)],
    "B": [("C", 2), ("D", 5)],
    "C": [("D", 1)],
    "D": []
}

heuristic = {"A": 4, "B": 3, "C": 1, "D": 0}

def astar(start, goal):
    pq = [(heuristic[start], 0, start, [start])]
    best = {start: 0}
```

```
while pq:
    f, cost, node, path = heapq.heappop(pq)

    if node == goal:
        print("Path:", " -> ".join(path))
        print("Cost:", cost)
        return

    if cost != best[node]:
        continue

    for neighbor, weight in graph[node]:
        new_cost = cost + weight

        if new_cost < best.get(neighbor, float("inf")):
            best[neighbor] = new_cost
            heapq.heappush(
                pq,
                (new_cost + heuristic[neighbor],
```

```
                new_cost, neighbor, path + [neighbor])
            )

astar("A", "D")
```

Output:

```
Path: A -> B -> C -> D
Cost: 4
```